Part E | Design justification

Channel budget

The question chooses the chart. GDP and healthcare access receive the two position channels because their relationship needs the most precise comparison; GDP is logarithmic and access keeps its 0–100 bounds. Population receives area, with an explicit affine 24–650 pt² range and a three-value legend. This makes small countries readable but compresses population ratios, so the data table provides exact values. Six region hues carry identity, not magnitude. The palette audit and accessible table disclose the limits of hue discrimination. Aldonia is retained and labelled; no country is removed and no Other bucket is needed.

Parallel coordinates

Percentile ranks prevent one extreme GDP observation from compressing the other 59 countries, at the cost of losing gap sizes. The greedy absolute-Pearson-correlation rule fixes the order before inspection: healthcare access, case rate, urban share, median age, GDP. Grey context and orange Andara provide one pop-out per chart. Access–incidence crossings reveal the inverse relationship; incidence–urbanisation also shows an inverse association (r=-0.619). Complete paths also identify the same Andara countries with above-median access and urban share plus below-median incidence; the notebook prints their names. An unlinked scatter matrix does not identify that same-row three-way conjunction without additional linking.

Spatial decisions

Cases divided by resident population times 100,000 estimates period incidence, while a companion zero-based sorted count bar describes workload. Five Jenks classes maximise the reported within-class variance fit among the three tested schemes; printed numeric breaks expose the decision. They are descriptive, not clinical thresholds. Lambert cylindrical equal-area uses one shared reference latitude, preserving area across districts. All 24 records join without failure. Boundaries and data are synthetic CDB601220 Week 5 materials; the source does not specify a boundary year or observation period. Access date: 28 October 2026. No real-world year is invented.

Counterfactual and ethics

A raw-count choropleth would point at D17, D13 and D09, whereas the largest rates occur in D19, D20 and D21. Repeatedly colouring populous districts dark can stigmatise residents without establishing greater individual risk. Counts remain necessary for staffing, so both views are retained. Part C has zero supplied real-reader responses: its mean overlap is undefined and no behavioural claim is made. The deterministic top-three overlap of zero illustrates how the rankings differ; it is not experimental evidence. Publication suppresses district counts below five; none meet that rule. All 40 small-area units are retained only in the explicitly synthetic diagnostic audit, where instability is shown. A real release would need a context-specific disclosure assessment and aggregation of sensitive small counts, accepting loss of geographic detail.